

Exact Execution Guide to Complete the FR3 CBF Project

Standalone project package

23 July 2026

Immediate scientific task. Close the fixed-service S1 adequacy gate on real paired links. Do not spend the next work cycle adding new conceptual material. The first evidence must decide whether the provisional +19 dB short-term engineering candidate is acceptable, must be reduced, must be link-specific, or should not be used.

1 Day-zero software setup

From the package root:

```
python -m venv .venv
source .venv/bin/activate
# Windows PowerShell: .venv\Scripts\Activate.ps1
python -m pip install --upgrade pip
python -m pip install -r requirements-core.txt
python -m pip install -e . --no-build-isolation
python scripts/00_validate_package.py
python -m pytest
python scripts/01_run_smoke_test.py
```

Pass condition: package validation and tests pass. Demo results remain clearly marked as demo evidence.

2 Obtain the P.530-19 digital products

The included ITU archive contains the Recommendation PDF. The official numerical grids are a separate components ZIP.

Online:

```
python scripts/02_download_p530_products.py
```

Offline/manual:

1. Download English *Zip (Components)* from the official P.530-19 page.
2. Save it as `data/external/p530/P530_components.zip`.
3. Run the downloader with `-zip`.

Pass condition: LogK.csv, dN75.csv, coordinate products, and a SHA-256 manifest exist; the grid loader passes.

3 Create reviewed paired fixed links

A fixed receiver alone is not enough. S1 needs its wanted transmitter and link budget.

Use an explicit trusted pair key. The strict pairing script refuses nearest-neighbour or frequency-only inference:

```
python scripts/03a_pair_fs_records.py \
  --input <raw.csv> \
  --mapping config/tafl_column_mapping.yaml \
  --output data/real/paired_fs_links_unfinished.csv
```

Complete, for every link:

- endpoints and antenna altitudes above mean sea level;
- mean terrain elevation along the wanted path;
- frequency and documented fade margin;
- incremental worst-month outage allocation;
- source IDs and provenance.

Optional DEM helper:

```
python -m pip install -r requirements-full.txt
python scripts/03b_add_dem_mean_terrain.py \
  --links <paired.csv> --dem <dem.tif> \
  --output data/real/paired_fs_links_with_terrain.csv
```

Optional fade-margin helper, only when received level and receiver threshold are supplied:

```
python scripts/03c_compute_fade_margin.py \
  --input <links.csv> --output data/real/paired_fs_links.csv
```

4 Make the human engineering decision that code cannot make

The incremental-outage allocation is not guessed. Use `data/templates/S1_ALLOCATION_DECISION_TEMPLATE.md`. State the reference period, link population, value, rationale, and reviewer.

Then validate:

```
python scripts/03_validate_paired_links.py \
  data/real/paired_fs_links.csv --mode real
```

Pass condition: no missing numeric field, duplicate ID, example link, empty provenance, or geometry error.

5 Run nominal S1 and sensitivity

Edit `config/s1_adequacy.yaml` so real mode points to the reviewed file and the allocation is explicit.

```
python scripts/04_run_s1_adequacy.py \
```

```
--config config/s1_adequacy.yaml --mode real
python scripts/04b_run_s1_sensitivity.py \
--config config/s1_adequacy.yaml --mode real
```

Review the link-level table before the summary. A candidate passes only if every intended link meets its allocation and every predeclared sensitivity scenario remains acceptable.

The script evaluates candidate caps 10, 13, 16, 19, and 23 dB and reports baseline/post-interference P.530 worst-month outage, fade-margin penalty, incremental outage, warnings, and provenance.

6 Freeze, reduce, or reject the candidate

Only after review:

```
python scripts/05_freeze_s1_candidate.py \
--candidate-db 19 \
--reviewer "Name" \
--decision-note "Detailed reviewed rationale" \
--confirm
```

The script refuses demo evidence, a nominal failure, or a sensitivity failure. The output is a separate engineering-decision YAML. It does not silently rewrite the regulatory source file.

7 Build the paper experiments

7.1 E1: static fixed-service continuity

1. Generate real UMa cellular channels and nominal WMMSE beams in the existing simulator.
2. Generate or export validated P.452-18 coupling to the real FS receivers.
3. Export the NPZ bundle defined in `INTEGRATION_CONTRACT.md`.
4. Validate and run E1.
5. Confirm unit/bandwidth/array-gain accounting.

7.2 E2: dynamic fixed-service stress

1. Select a physical disturbance and the real rate-limited control variable.
2. Precompute time-varying robust coupling and one-step bounds.
3. Use identical information and slew limits for all methods.
4. Demonstrate the rate-limit trap.
5. Report trajectories, violations, rate, slack, control activity, and runtime.

7.3 E3: drifting earth-station geometry

1. Archive station assumptions, antenna pattern, and TLE.
2. Generate/validate the topocentric track.
3. Build time-varying sector-to-station coupling.
4. Enforce the long SA.1027 entry through the exceedance budget and the short entry as an always-on cap.
5. Repeat over passes, satellites, traffic seeds, and uncertainty levels.

8 Calibrate uncertainty and prove practicality

Use `scripts/07_run_calibration.py` on held-out residual data. Report overall and stress-stratified coverage. Use `scripts/16_benchmark_runtime.py` only as a reference starting point; final runtime must measure the real central benchmark and practical layered controller at paper scale.

9 Final completion gate

Run:

```
python scripts/15_check_submission_readiness.py
python scripts/14_generate_manifest.py --include-results
```

The paper is not ready while any required real audit, S1 decision, calibration result, runtime result, provenance file, or open standards/model item is missing.

10 Minimal weekly order

Order	Work	Output
1	Environment, tests, products	Clean reproducible package
2	Pair fixed links and fill terrain/fade margins	Reviewed S1 input
3	Approve outage allocation	Decision memo
4	Nominal and sensitivity S1	Candidate decision
5	Real E1 bundle	Static continuity results
6	E2 dynamic coupling and runs	Main CBF counterexample
7	E3 TLE/station/coupling and runs	Drifting-geometry results
8	Calibration and runtime	Coverage/practicality evidence
9	Statistics and paper population	Submission archive